

Power BI Data Modeling & Reporting Project

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Introduction

This project demonstrates the implementation of a structured data model using the star schema approach in Power BI. The objective is to create an efficient, scalable, and analysis-ready data model.

Objective

- Build a star schema model
- Establish correct relationships
- Enable hierarchical analysis
- Validate using report visuals

Data Model Overview

The model consists of a central Sales_Fact table connected to Customer, Product, Region, and Date dimension tables. A secondary Returns_Fact table is also included.

Relationships

All relationships follow many-to-one cardinality with single direction filtering. The Returns_Fact to Date_Dim relationship is inactive to prevent ambiguity.

Hierarchies

Date: Year > Quarter > Month > Date

Region: Country > State > City

Product: Category > Subcategory > Product Name

Report Design

The report includes KPI cards, category-wise revenue analysis, region-wise comparison, and time-based trends.

Verification

A matrix visual is used to validate the relationships and confirm accurate data flow across tables.

Conclusion

The project successfully implements a clean and efficient star schema model with a functional report. It follows industry best practices for Power BI data modeling.